Pentagonal Number Theorem

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1 Partitions

Definition 1. A *partition* of a positive integer n, also called an integer partition, is a way of writing n as a sum of positive integers. Two sums that differ only in the order of their summands are considered the same partition.

The *partition function* $p(n)$ represents the number of possible partitions of a natural number n.

Example 2. $5 = 4+1 = 3+2 = 3+1+1 = 2+2+1 = 2+1+1+1 = 1+1+1+1+1$
$p(5) = 7$

Lemma 3. *The generating function for $p(n)$ is given by*

$$\sum_{n=0}^{\infty} p(n)\, x^n = \prod_{k=1}^{\infty}(1 + x^k + x^{2k} + \ldots) = \prod_{k=1}^{\infty}(1 - x^k)^{-1}.$$

Proof. We work in the ring of formal power series $\mathbb{Z}[[x]]$.

Step 1. For each fixed $k \geq 1$, the geometric series identity

$$1 + x^k + x^{2k} + \cdots = \sum_{j=0}^{\infty} x^{jk} = (1 - x^k)^{-1}$$

holds as formal power series. This gives the second equality.

Step 2. Expand the product $\prod_{k=1}^{\infty}(1+x^k+x^{2k}+\cdots)$. Each term in the expansion is obtained by selecting, for every $k \geq 1$, an exponent $j_k \in \mathbb{Z}_{\geq 0}$ (with $j_k = 0$ for all but finitely many k) and multiplying the chosen monomials. Such a selection contributes a single term

$$\prod_k x^{j_k \cdot k} = x^{\sum_k j_k \cdot k}.$$

Step 3. The coefficient of x^n is therefore the number of tuples $(j_1, j_2, \ldots)$ with $\sum_k k \cdot j_k = n$.

Step 4. Such a tuple corresponds bijectively to a partition of n: j_k is the multiplicity of the part k. Hence the coefficient of x^n equals $p(n)$, and the first equality follows. □

Definition 4 (Partitions of n into distinct parts)**.** Let $n \in \mathbb{N}$. We define the following sets of partitions of n into distinct positive parts:

$$\mathcal{P}(n) = \left\{ S \subseteq \{1, \ldots, n\} \;\middle|\; \sum_{s \in S} s = n \right\},$$
$$\mathcal{P}_{\text{even}}(n) = \{S \in \mathcal{P}(n) \mid |S| \equiv 0 \pmod 2\},$$
$$\mathcal{P}_{\text{odd}}(n) = \{S \in \mathcal{P}(n) \mid |S| \equiv 1 \pmod 2\}.$$

We further define the number of even and odd partitions of n as $p_e(n) = |\mathcal{P}_{\text{even}}(n)|$ and $p_o(n) = |\mathcal{P}_{\text{odd}}(n)|$.

Lemma 5. *We have*

$$\prod_{k=1}^{\infty}(1-x^k) \;=\; \sum_{s=0}^{\infty} \sum_{k_1<k_2<\cdots<k_s} (-1)^s\, x^{k_1+k_2+\ldots+k_s} \;=\; \sum_{n=0}^{\infty} c_n\, x^n$$

where $c_0 = 1$ *and for* $n \geq 1$ *and* $c_n = p_e(n) - p_o(n)$.

Proof. We expand the infinite product as a formal power series.

Step 1: Expansion as a sum over finite subsets. For each $k \geq 1$, the factor $(1-x^k)$ contributes either 1 or $-x^k$ when expanding the product. A choice across all factors corresponds to a finite subset

$$K = \{k_1 < k_2 < \cdots < k_s\} \subseteq \mathbb{Z}_{\geq 1}$$

(the indices for which $-x^k$ was selected). The contribution is

$$\prod_{k\in K}(-x^k) = (-1)^{|K|}\, x^{\sum_{k\in K} k} = (-1)^s\, x^{k_1+\cdots+k_s}.$$

Summing over all such K yields the first equality.

Step 2: Grouping by total exponent. A subset K of size s summing to n is exactly a partition $S \in \mathcal{P}(n)$ with $|S| = s$. Hence the coefficient of x^n equals

$$c_n = \sum_{S\in\mathcal{P}(n)} (-1)^{|S|}.$$

Step 3: $c_0 = 1$. Only $S = \emptyset$ partitions 0 into distinct positive parts, contributing $(-1)^0 = 1$.

Step 4: $c_n = p_e(n) - p_o(n)$ for $n \geq 1$. Splitting the sum by parity of $|S|$:

$$c_n = \sum_{\substack{S\in\mathcal{P}(n)\\ |S| \text{ even}}} 1 \;-\; \sum_{\substack{S\in\mathcal{P}(n)\\ |S| \text{ odd}}} 1 = |\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)| = p_e(n) - p_o(n).$$

$\square$

Example 6. *We compute* $\prod_{i=1}^{20}(1-x^i) + o(x^{20})$.

The answer is $1 - x - x^2 + x^5 + x^7 - x^{12} - x^{15} + O\left(x^{20}\right)$.

n		$p_o(n)$	$p_e(n)$
1	1	1	0
2	2	1	0
3	$3 = 2+1$	1	1
4	$4 = 3+1$	1	1
5	$5 = 4+1 = 3+2$	1	2
6	$6 = 5+1 = 4+2 = 3+2+1$	2	2
7	$7 = 6+1 = 5+2 = 4+3 = 4+2+1$	2	3

2 Pentagonal number theorem

Theorem 7 (Euler)**.**

$$\prod_{i=1}^{\infty}(1-x^i) \;=\; 1+\sum_{k=1}^{\infty}(-1)^k\,(x^{(3k^2-k)/2}+x^{(3k^2+k)/2}) \;=\; \sum_{k\in\mathbb{Z}}(-1)^k\,x^{(3k^2-k)/2}.$$

Proof. The proof combines the previous expansion lemma with the closed-form computation of $p_e(n)-p_o(n)$ via Franklin's involution.

Step 1: Coefficient identification. By the previous lemma,

$$\prod_{i=1}^{\infty}(1-x^i)=\sum_{n=0}^{\infty}c_n\,x^n,$$

where $c_0=1$ and $c_n=p_e(n)-p_o(n)$ for $n\geq 1$.

Step 2: Closed form for c_n. By Lemma 24 below,

$$p_e(n)-p_o(n)=\begin{cases}(-1)^k & \text{if } n=(3k^2-k)/2 \text{ for some } k\in\mathbb{Z}_{\geq 1},\\ (-1)^k & \text{if } n=(3k^2+k)/2 \text{ for some } k\in\mathbb{Z}_{\geq 1},\\ 0 & \text{otherwise.}\end{cases}$$

Step 3: Reassembling the sum. The values of n where $c_n\neq 0$ are exactly the (generalized) pentagonal numbers $n=(3k^2\pm k)/2$ for $k\geq 1$, plus $n=0$. Each contributes $(-1)^k$. Hence

$$\prod_{i=1}^{\infty}(1-x^i)=1+\sum_{k=1}^{\infty}(-1)^k\left(x^{(3k^2-k)/2}+x^{(3k^2+k)/2}\right).$$

Step 4: Bilateral sum reformulation. Reindex the second family by $k\mapsto -k$. For $k\in\mathbb{Z}_{\geq 1}$, set $k'=-k$, so $k'\in\mathbb{Z}_{\leq -1}$ and

$$\frac{3(k')^2-k'}{2}=\frac{3k^2+k}{2},\qquad (-1)^k=(-1)^{|k'|}=(-1)^{k'}.$$

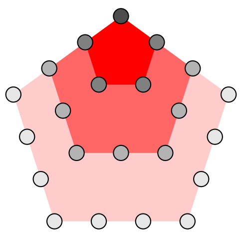
The $k=0$ term gives $(-1)^0\,x^0=1$, matching the leading 1. Combining the three contributions:

$$\sum_{k\in\mathbb{Z}_{\geq 1}}(-1)^k x^{(3k^2-k)/2}\;+\;(-1)^0x^0\;+\;\sum_{k'\in\mathbb{Z}_{\leq -1}}(-1)^{k'}x^{(3(k')^2-k')/2}\;=\;\sum_{k\in\mathbb{Z}}(-1)^k x^{(3k^2-k)/2}.$$

□

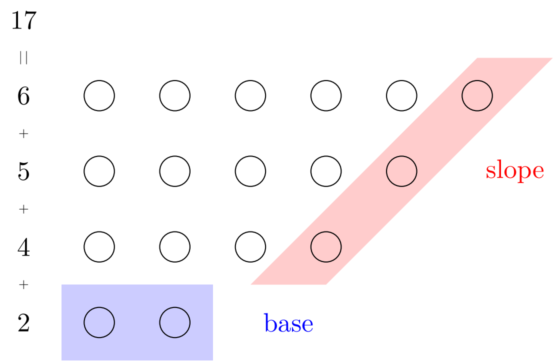
Remark 8 (Pentagonal numbers)**.**

k	*-3*	*-2*	*-1*	*0*	*1*	*2*	*3*
$\frac{3k^2-k}{2}$	*15*	*7*	*2*	*0*	*1*	*5*	*12*



2.1 Proof

Consider a diagram of a partition of n into different parts.



Definition 9 (Base and slope)**.** Consider a partition $S \subseteq \{1, \dots, n\}$ of n into different parts. We define the *base* $b \in S$ as the smallest element of the partition. Let $\max(S)$ denote the largest element of S. We define the *slope* s as:

$$s = \max\{k \geq 1 : \max(S), \max(S) - 1, \max(S) - 2, \dots, \max(S) - k + 1 \in S\}.$$

and the *slope set* D as $\{\max(S), \max(S) - 1, \dots, \max(S) - s + 1\}$.

Definition 10. Let $n \in \mathbb{N}$. We define the following sets of partitions of n into distinct positive parts:

$$\begin{aligned}
\mathcal{P}_\alpha(n) &= \{S \in \mathcal{P}(n) \mid b \leq s \text{ and } b \notin D, \text{ or if } b \leq s - 1\}, \\
\mathcal{P}_\beta(n) &= \{S \in \mathcal{P}(n) \mid b > s \text{ and } b \notin D, \text{ or if } b \geq s + 2\}, \\
\mathcal{P}_{\text{special}}(n) &= \{S \in \mathcal{P}(n) \mid b \in D \text{ and } |b| = |s| \text{ or } |b| = |s| + 1\}.
\end{aligned}$$

Lemma 11. *For all natural numbers n the following holds: The sets $\mathcal{P}_\alpha(n)$, $\mathcal{P}_\beta(n)$, $\mathcal{P}_{\text{special}}(n)$ are pairwise disjoint.*

Proof. Throughout, let $S \in \mathcal{P}(n)$ be a partition with base b, slope s, slope set D. Recall the membership criteria:

- $S \in \mathcal{P}_\alpha$ iff ($b \leq s$ and $b \notin D$) or $b \leq s - 1$;
- $S \in \mathcal{P}_\beta$ iff ($b > s$ and $b \notin D$) or $b \geq s + 2$;
- $S \in \mathcal{P}_{\text{special}}$ iff $b \in D$ and ($b = s$ or $b = s + 1$).

Step 1: $\mathcal{P}_\alpha \cap \mathcal{P}_\beta = \emptyset$. Suppose $S \in \mathcal{P}_\alpha \cap \mathcal{P}_\beta$. Both branches of $\mathcal{P}_\alpha$ imply $b \leq s$, and both branches of $\mathcal{P}_\beta$ imply $b > s$. Contradiction.

Step 2: $\mathcal{P}_\alpha \cap \mathcal{P}_{\text{special}} = \emptyset$. Suppose $S \in \mathcal{P}_{\text{special}}$, so $b \in D$ and $b \in \{s, s + 1\}$.

- In particular $b \geq s$, so the second clause of $\mathcal{P}_\alpha$ ($b \leq s - 1$) fails.
- The first clause requires $b \notin D$, but $b \in D$, so it also fails.

Hence $S \notin \mathcal{P}_\alpha$.

Step 3: $\mathcal{P}_\beta \cap \mathcal{P}_{\text{special}} = \emptyset$. Suppose $S \in \mathcal{P}_{\text{special}}$, so $b \in D$ and $b \leq s + 1$.

- The second clause of $\mathcal{P}_\beta$ ($b \geq s+2$) fails since $b \leq s+1$.
- The first clause requires $b \notin D$, contradicted by $b \in D$.

Hence $S \notin \mathcal{P}_\beta$. $\square$

Lemma 12. *For all natural numbers n the following holds*

$$\mathcal{P}(n) = \mathcal{P}_\alpha(n) \sqcup \mathcal{P}_\beta(n) \sqcup \mathcal{P}_{\text{special}}(n).$$

Proof. By the previous lemma the three sets are pairwise disjoint, so it suffices to show that every $S \in \mathcal{P}(n)$ belongs to at least one of them.

Step 1: Edge case. If $S = \emptyset$ (which can occur only when $n = 0$), the base, slope, and slope set are not defined. We treat this case separately: there are no special, α, or β partitions of 0 in the sense above, so the statement reduces to $\mathcal{P}(0) = \emptyset \sqcup \emptyset \sqcup \emptyset$. TODO: confirm/define the convention for $n = 0$ — the natural reading is $\mathcal{P}(0) = \{\emptyset\}$ but no operation applies. The statement either needs $n \geq 1$ or a convention placing $\emptyset$ in one of the three sets.

Step 2: Main case ($S \neq \emptyset$). Let $b = \min(S)$, s the slope, D the slope set. We perform a trichotomy on b vs s:

Case 1: $b \leq s - 1$. The second clause of $\mathcal{P}_\alpha$ holds, so $S \in \mathcal{P}_\alpha$.

Case 2: $b = s$.

- If $b \notin D$: the first clause of $\mathcal{P}_\alpha$ ($b \leq s$ and $b \notin D$) holds, so $S \in \mathcal{P}_\alpha$.
- If $b \in D$: then $b = s$ and $b \in D$, so $S \in \mathcal{P}_{\text{special}}$.

Case 3: $b = s + 1$.

- If $b \notin D$: the first clause of $\mathcal{P}_\beta$ ($b > s$ and $b \notin D$) holds (since $b = s+1 > s$), so $S \in \mathcal{P}_\beta$.
- If $b \in D$: then $b = s+1$ and $b \in D$, so $S \in \mathcal{P}_{\text{special}}$.

Case 4: $b \geq s + 2$. The second clause of $\mathcal{P}_\beta$ holds, so $S \in \mathcal{P}_\beta$.

These four cases cover all possibilities (either $b \leq s+1$ or $b \geq s+2$; within $b \leq s+1$ we split into $b \leq s-1$, $b = s$, $b = s+1$). $\square$

Definition 13. For $k \in \mathbb{Z}_{\geq 1}$ define $S_{-k} = \{2k-1, 2k-2, \dots, k\}$.
For $k \in \mathbb{Z}_{\geq 0}$ define $S_k = \{2k, 2k-1, \dots, k\}$.

Lemma 14. *If n is not a pentagonal number, then $\mathcal{P}_{\text{special}}(n) = \emptyset$.*
If $n = \frac{3k^2-k}{2}$ for some $k \in \mathbb{Z}_{\geq 1}$, then $\mathcal{P}_{\text{special}}(n) = \{S_{-k}\}$.
If $n = \frac{3k^2+k}{2}$ for some $k \in \mathbb{Z}_{\geq 0}$, then $\mathcal{P}_{\text{special}}(n) = \{S_k\}$.

Proof. TODO: The statement and the definition of S_k for the upper-staircase family require reconciliation. As written, $S_k = \{2k, 2k-1, \dots, k\}$ has $k+1$ elements summing to $\sum_{i=k}^{2k} i = (3k^2+3k)/2 \neq (3k^2+k)/2$. The intended set is $\{k+1, k+2, \dots, 2k\}$ with k elements (and base $k+1$, slope k). Also the case $k = 0$ is degenerate since it would require $S = \{0\}$, which is not a partition into positive parts. The statement should be: for $k \in \mathbb{Z}_{\geq 1}$, $\mathcal{P}_{\text{special}}((3k^2+k)/2) = \{\{k+1, k+2, \dots, 2k\}\}$. The proof below uses this corrected reading.

Step 1: Structure of a special partition. Let $S \in \mathcal{P}_{\text{special}}(n)$ with base b, slope s, slope set D. By definition, $b \in D$ and either $b = s$ or $b = s + 1$.

Sub-step 1a: $S = D$. The slope set is the maximal top-consecutive run: $D = \{\max(S) - s + 1, \dots, \max(S)\}$. Since $b \in D$ and $b = \min(S)$, every element of S lies between b and $\max(S)$.

Moreover, the integers $b, b+1, \ldots, \max(S)$ all lie in D (because $b \in D$ implies $b \geq \max(S) - s + 1$, so $\{b, b+1, \ldots, \max(S)\} \subseteq D$), and $D \subseteq S$. Combined with $b = \min(S)$, this forces

$$S = \{b, b+1, \ldots, \max(S)\}.$$

In particular $|S| = \max(S) - b + 1$, and since $S = D$ this also equals s. Thus $\max(S) = b + s - 1$.

Step 2: Case A — $b = s$. Then $|S| = s = b$ and $\max(S) = 2b - 1$, so

$$S = \{b, b+1, \ldots, 2b-1\}.$$

The sum is

$$\sum_{i=b}^{2b-1} i = b \cdot b + (0 + 1 + \cdots + (b-1)) = b^2 + \frac{b(b-1)}{2} = \frac{3b^2 - b}{2}.$$

Setting $k = b \in \mathbb{Z}_{\geq 1}$, we have $n = (3k^2 - k)/2$ and $S = S_{-k}$ (matching the definition $S_{-k} = \{2k-1, \ldots, k\}$).

Step 3: Case B — $b = s + 1$. Then $|S| = s = b - 1$ and $\max(S) = b + s - 1 = 2b - 2$, so

$$S = \{b, b+1, \ldots, 2b-2\}.$$

The sum is

$$\sum_{i=b}^{2b-2} i = (b-1) \cdot b + (0 + 1 + \cdots + (b-2)) = b(b-1) + \frac{(b-1)(b-2)}{2} = \frac{(b-1)(3b-2)}{2}.$$

Setting $k = b - 1 \in \mathbb{Z}_{\geq 1}$ (note $b \geq 2$ since $s \geq 1$), we get

$$n = \frac{k(3(k+1)-2)}{2} = \frac{k(3k+1)}{2} = \frac{3k^2+k}{2}, \qquad S = \{k+1, k+2, \ldots, 2k\}.$$

Step 4: Forward direction. Conversely, given $n = (3k^2 - k)/2$ with $k \geq 1$, the set $S_{-k} = \{k, k+1, \ldots, 2k-1\}$ has base k, slope k (the entire set is a top-consecutive run), and $b = k \in D$, hence is special. Similarly for the upper family.

Step 5: Uniqueness. The case analysis in Steps 2–3 shows that for each pentagonal n, there is at most one special partition (the candidate is uniquely determined by b). Combined with the forward direction, this gives exactly one.

Step 6: Non-pentagonal n. If n is not pentagonal of either form, no $b \geq 1$ produces $\sum = n$ in the analysis above, so $\mathcal{P}_{\text{special}}(n) = \emptyset$. □

Definition 15 (Operations α and β)**.** Let $S \subseteq \{1, \ldots, n\}$ be a partition of n into different parts with base b and slope s and slope set D. We define the following operations on S:

α: If $S \in \mathcal{P}_\alpha$ (equivalently, if $b \leq s$ and $b \notin D$, or if $b \leq s - 1$), then we move b up and turn its points into new slope.

$$\alpha(S) = (S \quad \{b, \max(S) - b + 1\}) \cup \{\max(S) + 1\}.$$

β: If $S \in \mathcal{P}_\beta$ (equivalently, if $b > s$ and $b \notin D$, or if $b \geq s + 2$), then we move the slope down and turn its points into the new base.

$$\beta(S) = S \cup \{s, \max(S) - s\} \quad \{\max(S)\}.$$

α

β

Lemma 16. *If $S \in \mathcal{P}_\alpha(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n)$.*

Proof. Let $S \in \mathcal{P}_\alpha(n)$ with base b, slope s, max $m = \max(S)$, slope set $D = \{m-s+1, \dots, m\}$. Recall that the α-condition says $b \le s$ (with $b \notin D$ if $b = s$), and

$$\alpha(S) = (S \setminus \{b, m-b+1\}) \cup \{m+1\}.$$

(Equivalently in the more granular form used in the Lean code: remove the base b, remove the bottom-most element of the top-b slope $\{m-b+1\}$, and add $m+1$. The remaining $b-1$ slope elements stay where they are; the smallest one of them is $m-b+2$.)

Step 1: $\alpha(S)$ is a partition of n. The change in sum is $-b-(m-b+1)+(m+1) = 0$, so the sum is preserved. Since b and $m-b+1$ both lie in S and $m+1 \notin S$ (as $m = \max(S)$), the operation produces a set, and all parts are positive (the smallest possible new value is $m+1 \ge 2$, and any retained part is in S, hence $\ge b \ge 1$).

We must verify that $b \ne m-b+1$ (so we are removing two distinct elements). Equivalently $m \ne 2b-1$. If $m = 2b-1$, then $S \supseteq \{b, b+1, \dots, 2b-1\} = D$, forcing $S = D$ (since $|S| \ge |D| = s = b$ and $S = \{b, \dots, 2b-1\}$). But then S would be the special partition S_{-b}, contradicting $S \in \mathcal{P}_\alpha$.

Step 2: New extremes.

- New maximum: $m' := \max(\alpha(S)) = m+1$.
- New base: $b' := \min(\alpha(S))$. Since b was removed, b' is the second-smallest element of S (if $|S| \ge 2$) or $m+1$ (if $|S| = 1$).

Step 3: New slope s' and slope set D'. The slope of $\alpha(S)$ is the maximal consecutive run from m' downward. The retained top-$(b-1)$ slope elements $\{m-b+2, \dots, m\}$ together with the new element $m+1$ form the consecutive run $\{m-b+2, m-b+3, \dots, m, m+1\}$ of length b. We claim this is the entire new slope, i.e., $s' = b$ and $D' = \{m-b+2, \dots, m+1\}$.

Sub-step 3a: The run does not extend further down. The next candidate would be $m-b+1$, which we just removed; so $m-b+1 \notin \alpha(S)$, terminating the run.

Step 4: Verify $\alpha(S) \in \mathcal{P}_\beta(n)$. We need $b' > s'$ (and $b' \notin D'$) or $b' \ge s'+2$. We have $s' = b$ from Step 3, and we examine which clause of $\mathcal{P}_\alpha$ produced S:

Sub-case (i): $b \le s-1$ (the second clause of $\mathcal{P}_\alpha$). The slope of S has length $s \ge b+1$, so its top b elements $\{m-b+1, \dots, m\}$ are a strict subset of D. The element $m-s+1 \in D$ satisfies $m-s+1 < m-b+1$, so $m-s+1$ is in S and below the part removed. After applying α, this element remains, and its value $m-s+1$ may or may not be the new minimum.

TODO: Case analysis on the location of the new base b' relative to D'. The careful analysis (matching the Lean lemma `alpha_gives_beta_crit`) shows: the new base $b' \geq b+1$ (since the second-smallest element of S is $\geq b+1$; in fact $\geq b+1$ because elements are distinct), and one verifies that $b' \geq s'+2$ or $b' > s' \wedge b' \notin D'$. The detailed bookkeeping requires distinguishing whether the second-smallest of S is in the slope or below it.

Sub-case (ii): $b = s$ and $b \notin D$ (first clause of $\mathcal{P}_\alpha$). TODO: Symmetric analysis. The conclusion is again that the resulting partition satisfies the β-criterion. □

Lemma 17. *If $S \in \mathcal{P}_\beta(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n)$.*

Proof. Let $S \in \mathcal{P}_\beta(n)$ with base b, slope s, max m, slope set D. Recall the β-condition: $b > s$ (with $b \notin D$ if $b = s+1$), and

$$\beta(S) = (S \cup \{s, m-s\}) \ \ \{m\}.$$

(In the granular form: remove the top-of-slope m, shift the slope $D = \{m-s+1, \dots, m\}$ down by one to $\{m-s, \dots, m-1\}$, and adjoin a new bottom element $\{s\}$. The retained slope elements stay at $\{m-s+1, \dots, m-1\}$; the only "new" element below them is $m-s$, and the only "new" element overall is s.)

Step 1: $\beta(S)$ is a partition of n. Sum change: shifting s slope elements each down by 1 contributes $-s$, and adjoining the new singleton $\{s\}$ contributes $+s$. The net change is zero.

We must verify the resulting set has distinct elements:

- $s \notin S \ \ \{m\}$: since $b > s$ (from β-criterion) and $b = \min(S)$, all parts of S exceed s. So s is genuinely new.
- $m - s \notin S \ \ \{m\}$: if $m-s$ were already in S, then since $m-s < m-s+1$ (the slope minimum), $m-s$ would be in $S \ \ D$. But by maximality of the slope (the slope is the maximal top-consecutive run; cf. `slope_is_interval` in Lean), if $m-s \in S$ then $\{m-s, m-s+1, \dots, m\} \subseteq S$, which would mean $m-s \in D$, contradiction.
- $s \neq m-s$, i.e., $m \neq 2s$. TODO: verify this from the β-criterion. Heuristically, if $m = 2s$ then $D = \{s+1, \dots, 2s\}$ has minimum $s+1$; combined with β-criterion ($b \geq s+2$ in the second clause, or $b = s+1, b \notin D$ in the first), one checks the case $m = 2s, b = s+1$ would give $b \in D$ (since $b = s+1 \in D$), excluded.
- Positivity: $s \geq 1$ and $m-s \geq 1$ (since $m-s \geq m-s+1-1$ and the slope minimum is ≥ 1).

Step 2: New extremes.

- New base: $b' := \min(\beta(S)) = \min(b, s) = s$ (since $b > s$).
- New maximum: $m' := \max(\beta(S))$. If $|S| \geq 2$, then $m' =$ second-largest element of S, which equals $m-1$ (the next slope element). If $|S| = 1$, then $S = \{m\} = \{b\}$, but $b > s \geq 1$ means $s \geq 1$, so $|S| = 1$ forces a degenerate case; TODO: verify $|S| \geq 2$ holds whenever $S \in \mathcal{P}_\beta$, by a small computation.

Step 3: New slope s' and slope set D'. The shifted slope occupies $\{m-s, m-s+1, \dots, m-1\}$, a consecutive run of length s ending at $m' = m-1$. So the new slope is at least this run. We claim it is exactly this run, i.e., $s' = s$ and $D' = \{m-s, \dots, m-1\}$.

Reason it does not extend further down: the next candidate is $m-s-1$. If this were in $\beta(S)$, it would have to come from $S \ \ \{m\}$ (since neither shift-target nor s equals $m-s-1$ generically).

But $m-s-1 \in S$ would, by slope maximality applied to S, require $m-s-1 \in D$, contradicting $\min(D) = m-s+1$.

Step 4: Verify $\beta(S) \in \mathcal{P}_\alpha(n)$. We have $b' = s$ and $s' = s$, so $b' = s'$. We need either $b' \notin D'$ (giving the first clause of $\mathcal{P}_\alpha$) or $b' \leq s'-1$ (the second clause). Since $b' = s'$, the second clause fails, and we must show $b' = s \notin D'$.

$D' = \{m-s, m-s+1, \dots, m-1\}$. We have $s \in D'$ iff $m-s \leq s \leq m-1$, iff $m \geq s+1$ (always true since $m \geq b > s$) and $s \leq m-1$, the latter equivalent to $m \geq s+1$.

So generically $s \in D'$ would hold, which would block $\mathcal{P}_\alpha$. But we have additional information from the β-criterion:

- If $b \geq s+2$ (second clause of β): then $m \geq b \geq s+2$, so $m-s \geq 2 > s$ would require... TODO: this analysis is delicate and the careful statement is that $s \notin D'$, which translates to $s < m-s$, i.e., $m > 2s$. This holds in both clauses of $\mathcal{P}_\beta$, but needs the same kind of casework as in the α direction. See Lean's `beta_gives_alpha_crit` (currently sorry-ed).

□

Lemma 18. *The composition of maps $\beta \circ \alpha$ is equal to the identity map on $\mathcal{P}_\alpha(n)$. Equivalently, if $S \in \mathcal{P}_\alpha(n)$ then $\beta(\alpha(S)) = S$.*

Proof. Let $S \in \mathcal{P}_\alpha(n)$ with base b, slope s, max m, slope set D.

Step 1: Compute $\alpha(S)$. By the definition of α:

$$\alpha(S) = (S \quad \{b, m-b+1\}) \cup \{m+1\}.$$

By Lemma 16, $\alpha(S) \in \mathcal{P}_\beta(n)$. From the proof of that lemma, the data of $\alpha(S)$ is:

- new max $m^\alpha = m+1$,
- new slope $s^\alpha = b$ (the run $\{m-b+2, \dots, m+1\}$),
- new slope set $D^\alpha = \{m-b+2, \dots, m+1\}$.

Step 2: Apply β to $\alpha(S)$. By definition of β:

$$\beta(\alpha(S)) = (\alpha(S) \cup \{s^\alpha, m^\alpha - s^\alpha\}) \quad \{m^\alpha\} = (\alpha(S) \cup \{b, m+1-b\}) \quad \{m+1\}.$$

Step 3: Simplify. Substituting $\alpha(S) = (S \quad \{b, m-b+1\}) \cup \{m+1\}$:

$$\begin{aligned}
\beta(\alpha(S)) &= [((S \quad \{b, m-b+1\}) \cup \{m+1\}) \cup \{b, m-b+1\}] \quad \{m+1\} \\
&= [(S \quad \{b, m-b+1\}) \cup \{b, m-b+1, m+1\}] \quad \{m+1\} \\
&= (S \quad \{b, m-b+1\}) \cup \{b, m-b+1\} \\
&= S,
\end{aligned}$$

where we used that b and $m-b+1$ are in S (so re-adding them after removal recovers S), and that $m+1 \notin S \quad \{b, m-b+1\}$ (since $m+1 > m \geq$ every element of S). □

Lemma 19. *The composition of maps $\alpha \circ \beta$ is equal to the identity map on $\mathcal{P}_\beta(n)$. Equivalently, if $S \in \mathcal{P}_\beta(n)$ then $\alpha(\beta(S)) = S$.*

Proof. Let $S \in \mathcal{P}_\beta(n)$ with base b, slope s, max m, slope set D.

Step 1: Compute $\beta(S)$.

$$\beta(S) = (S \cup \{s, m-s\}) \quad \{m\}.$$

By Lemma 17, $\beta(S) \in \mathcal{P}_\alpha(n)$. From the proof of that lemma, the data of $\beta(S)$ is:

- new max $m^\beta = m - 1$,
- new base $b^\beta = s$,
- new slope $s^\beta = s$ (the run $\{m-s, \dots, m-1\}$),
- new slope set $D^\beta = \{m-s, \dots, m-1\}$.

Step 2: Apply α to $\beta(S)$.

$$\alpha(\beta(S)) = (\beta(S)\ \ \{b^\beta, m^\beta - b^\beta + 1\}) \cup \{m^\beta + 1\} = (\beta(S)\ \ \{s, m-s\}) \cup \{m\}.$$

(We used $b^\beta = s$, $m^\beta = m-1$, so $m^\beta - b^\beta + 1 = m - 1 - s + 1 = m - s$.)

Step 3: Simplify. Substituting $\beta(S) = (S \cup \{s, m-s\})\ \ \{m\}$:

$$\begin{aligned} \alpha(\beta(S)) &= [((S \cup \{s, m-s\})\ \ \{m\})\ \ \{s, m-s\}] \cup \{m\} \\ &= [(S\ \ \{m\})\ \ \{s, m-s\}] \cup \{m\}. \end{aligned}$$

Since $s \notin S$ and $m - s \notin S$ (verified in the proof of Lemma 17), removing them from $S\ \ \{m\}$ has no effect:

$$\alpha(\beta(S)) = (S\ \ \{m\}) \cup \{m\} = S,$$

since $m \in S$. □

Lemma 20. *The map $\alpha : \mathcal{P}_\alpha(n) \to \mathcal{P}_\beta(n)$ is a bijection.*
The map $\beta : \mathcal{P}_\beta(n) \to \mathcal{P}_\alpha(n)$ is a bijection.

Proof. Lemma 16 shows α is well-defined as a map $\mathcal{P}_\alpha(n) \to \mathcal{P}_\beta(n)$, and Lemma 17 shows β is well-defined as a map $\mathcal{P}_\beta(n) \to \mathcal{P}_\alpha(n)$.

Lemma 18 gives $\beta \circ \alpha = \mathrm{id}_{\mathcal{P}_\alpha(n)}$, and Lemma 19 gives $\alpha \circ \beta = \mathrm{id}_{\mathcal{P}_\beta(n)}$.

Hence α and β are mutually inverse bijections. □

Lemma 21.

$$|\mathcal{P}_\alpha(n)| = \left|\mathcal{P}_\beta(n)\right|.$$

Proof. A bijection between finite sets implies equal cardinality. Apply this to the bijection $\alpha : \mathcal{P}_\alpha(n) \to \mathcal{P}_\beta(n)$ from Lemma 20. □

Lemma 22. *The operations α and β change the parity of the number of parts of a partition. More precisely,*

- *If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\mathrm{odd}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\mathrm{even}}(n)$;*
- *If $S \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\mathrm{even}}(n)$ then $\alpha(S) \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\mathrm{odd}}(n)$;*
- *If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\mathrm{odd}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\mathrm{even}}(n)$;*
- *If $S \in \mathcal{P}_\beta(n) \cap \mathcal{P}_{\mathrm{even}}(n)$ then $\beta(S) \in \mathcal{P}_\alpha(n) \cap \mathcal{P}_{\mathrm{odd}}(n)$.*

Proof. Membership in $\mathcal{P}_\beta$ for $\alpha(S)$ (and in $\mathcal{P}_\alpha$ for $\beta(S)$) is supplied by Lemmas 16 and 17. It remains to show that the parity of the number of parts is flipped.

Step 1: $|\alpha(S)| = |S| - 1$. By the definition $\alpha(S) = (S\ \ \{b, m-b+1\}) \cup \{m+1\}$:

- we remove two distinct elements (b and $m - b + 1$, distinct as shown in the proof of Lemma 16), reducing the size by 2;

- we add one element $(m+1)$, which is not in $S \quad \{b, m-b+1\}$ since $m+1 > \max(S)$.

Net change: $-2+1=-1$. So $|\alpha(S)| = |S|-1$, flipping parity.
Step 2: $|\beta(S)| = |S|+1$. By the definition $\beta(S) = (S \cup \{s, m-s\}) \quad \{m\}$:

- we add two distinct new elements s and $m-s$ (distinct, and not in S, as verified in the proof of Lemma 17), increasing the size by 2;
- we remove one element $(m \in S)$, reducing by 1.

Net change: $+2-1=+1$. So $|\beta(S)| = |S|+1$, flipping parity.
Step 3: Conclusion.

- $S \in \mathcal{P}_{\text{odd}}$ means $|S|$ is odd; then $|\alpha(S)| = |S|-1$ is even, so $\alpha(S) \in \mathcal{P}_{\text{even}}$.
- $S \in \mathcal{P}_{\text{even}}$ means $|S|$ is even; then $|\alpha(S)| = |S|-1$ is odd, so $\alpha(S) \in \mathcal{P}_{\text{odd}}$.
- Symmetrically for β with $|\beta(S)| = |S|+1$.

$\square$

Lemma 23.

$$|\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{odd}}(n)| = \left|\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{even}}(n)\right|,$$
$$|\mathcal{P}_\alpha(n) \cap \mathcal{P}_{\text{even}}(n)| = \left|\mathcal{P}_\beta(n) \cap \mathcal{P}_{\text{odd}}(n)\right|.$$

Lemma 24.

$$p_e(n) - p_o(n) = \begin{cases} (-1)^k, & \textit{if } n = \frac{3k^2 \pm k}{2} \\ 0, & \textit{otherwise.} \end{cases}$$

Proof. We have
$$p_e(n) - p_o(n) = |\mathcal{P}_{even}(n)| - |\mathcal{P}_{odd}(n)|. \tag{1}$$
By Lemma 12 we have
$$|\mathcal{P}_{\text{even}}(n)| = |\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\alpha(n)| + \left|\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_\beta(n)\right| + \left|\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)\right|$$
and
$$|\mathcal{P}_{\text{odd}}(n)| = |\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_\alpha(n)| + \left|\mathcal{P}_{\text{odd}}(n) \cap \mathcal{P}_\beta(n)\right| + \left|\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)\right|.$$
It follows from two previous equations and Lemma 23 that
$$|\mathcal{P}_{\text{even}}(n)| - |\mathcal{P}_{\text{odd}}(n)| = \left|\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)\right| - \left|\mathcal{P}_{\text{even}}(n) \cap \mathcal{P}_{\text{special}}(n)\right|. \tag{2}$$
Finally, the statement of the lemma follows from the equations (1), (2) and Lemma 14. $\square$

3 Jacobi triple product

Theorem 25. *For all $w \neq 0$ and q with $|q| < 1$:*

$$\prod_{k=1}^{\infty} (1-q^{2k})\,(1-q^{2k-1}\,w^2)\,(1-q^{2k-1}\,w^{-2}) = \sum_{k=-\infty}^{\infty} (-1)^k\, w^{2k}\, q^{k^2}.$$

3.1 Proof

Set $\phi_n(w,q) := \prod_{k=1}^{n}(1-q^{2k-1}\,w^2)\,(1-q^{2k-1}\,w^{-2})$. We have

$$\phi_n(qw,q) \;=\; \phi_n(w,q)\,\frac{(1-q^{2n+1}\,w^2)\,(1-q^{-1}w^{-2})}{(1-qw^2)\,(1-q^{2n-1}w^{-2})} \;=\; \phi_n(w,q)\,\frac{1-q^{2n+1}\,w^2}{-qw^2+q^{2n}}. \tag{3}$$

We consider the following expansion

$$\phi_n(w,q) = \sum_{k=-n}^{n} A_{n,k}(q)\,w^{2k}, \tag{4}$$

where $A_{n,k}(q)$ is a polynomial in q. We have a symmetry $A_{n,k}(q) = A_{n,-k}(q)$. We find

$$A_{n,n}(q) = q^{1+3+5+\ldots+(2n-1)} = q^{n^2}. \tag{5}$$

We substitute (4) into (3) and obtain

$$A_{n,k}(q) = A_{n,k-1}(q)\,q^{2k-1}\,\frac{1-q^{2n-2k+2}}{1-q^{2n+2k}}. \tag{6}$$

From (5) and (6) we find

$$A_{n,k} = \frac{q^{k^2}}{(1-q^2)\cdots(1-q^{2n})}\prod_{s=n-k+1}^{n}(1-q^{2s})\prod_{s=n+k+1}^{2n}(1-q^{2s}). \tag{7}$$

From (7) we obtain

$$\prod_{k=1}^{n}(1-q^{2k})\,(1-q^{2k-1}\,w^2)\,(1-q^{2k-1}\,w^{-2}) = \sum_{k=-n}^{n}(-1)^k w^{2k}\,q^{k^2}\,B_{n,k}(q).$$

where

$$B_{n,k}(q) = \prod_{s=n-k+1}^{n}(1-q^{2s})\prod_{s=n+k+1}^{2n}(1-q^{2s}).$$

We have $B_{n,k} = 1 + O(q^{n-k+1})$. We take the limit $n \to \infty$ and finish the proof.